ANALYTICS SPECIALIZATIONS & APPLICATIONS, 2024/25

COURSEWORK 1 - Customer Analytics Study

Coursework Release Date: Thursday, 30th January 2026
Coursework Deadline Date: Thursday, 5th March 2026, 3pm
Coursework Submission: via Moodle coursework submission link on the ASA web page

Work Plan Due: Friday 20th February 2026, 3pm
Work Plan Submission: via email to georgiana.nica-avram1@nottingham.ac.uk

Q&A Session: Friday 13th February 2026, 10am

1. The Problem Definition:

In this coursework, you will perform a market segmentation on a transactional dataset that has been provided by a national convenience store chain (4 files describing 3000 customers over 6 months).

→ Through analysis of the company's point-of-sale data you will produce profiles for 5-7 customer segments.

→ These will include a statistical summary and a pen profile for each segment (these would be used by the retailer to better understand their customer-base and to inform future marketing campaigns).

→ Your final deliverable in this task will be a report angled as though it is to be read by the company's chief data officer and marketing director (and so which should be presented accordingly).

→ This report will be accompanied by your technical implementation (expected to be in the form of Python and/or SQL scripts) and a .csv file that lists the customers assignments into the categories you have generated (simply linking their customer IDs to a segment id as indicated in your report).

2. Expected Approach:

Being a subjective process, this customer segmentation study has been deliberately left relatively open-ended. There are numerous ways of approaching a task of this sort and the key is to simply justify the method you take. Nevertheless, it is broadly expected that you will implement your segmentation study in three parts:

1. Exploration of the sample data provided to establish which indicators you believe are important to describe customer behaviour (justifying your decisions);
2. A quantitative analysis, considering the customer base in light of selected/ engineered features before then implementing a clustering approach to settle upon 5-7 customer segments (justifying your chosen methodology);

3. An analysis of results in order to produce a tangible description of each cluster in the form of a statistical summary, customer archetypes (pen profiles) and identification of the two most attractive targets for the company.

You should also submit a very short **bullet-pointed work plan (of no more than one page)** prior to implementation of the final task (deadline 3pm, Friday 20th February) and worth 5% of your submission. This brief bullet-pointed work plan should set out the form your analytical approach is expected to take, including a list of the features that you expect to use/engineer in order use to describe customers (whether RFM features, product-based spends, temporal features, etc.), your proposed technical strategy for doing the segmentation and brief details of how you intend to implement it (Note that this work plan is not binding, and your final report may differ from it).

3. Customer Data Provided

The company has asked you to directly use a sample of 3000 customers' behavioural data, as recorded by loyalty card transaction logs. You must use this data alone to provide the client a summarisation of their customer base in the form of 5-7 cluster archetypes - selected in a manner and number that you feel best describes the variation in the different types of customers the client interacts with. The files available to you are as follows:

FIle	Feature Description
customers_sample.csv	A summary file detailing the consumer behaviour of 3000 customers (referenced by an anonymised but consistent **"customer_id"**). This data details the total no. of "**baskets**" (i.e. visits) the customer has processed at the retailer's store over a 6 month period, the "**total_quantity"** of items they have purchased, the **"average_quantity"** per basket, the "**total_spend"** the customer has made over the period, and "**average_spend"** per visit.
category_ spends_sample.csv	This file again lists the 3000 customers in the sample, but this time splits down their spend over the period into 20 item categories, which represent the range of items sold across the retailer. The names of these categories are included in the file's header, and are self-explanatory.
baskets_sample.csv	This file details information about each individual visit made by the 3000 customers in the sample. Specifically ,it details the timestamp for the visit (**"purchase_time"**), quantity of items they purchased (**"basket_quantity"**), the amount of money they spent on that visit (**"basket_spend"**) and the number of different categories their purchases cut across for that trip (**"basket_categories"**)
lineitem_sample.csv,	A final dataset is also available for use that breaks down each basket into its individual product purchase ids, along with the category the item belongs to. This is provided only if you want to extend your analysis into new features, and all customer_ids it references correspond to those in the other data files.

3. Report Structure

You must provide a report that clearly describes your proposed customer segmentation of the client's consumer base, based on the sample of transactional data they have provided. The client company has requested that the number of groups to be generated be between 5 and 7, but that exact number is up to you. The customer segmentation process will require a stage of basic statistical analysis, a stage of feature selection, a stage of clustering, a stage of generating segments and then an analysis of the results to generate pen profiles - these must all be described (please take note of the marking scheme to decide how much of the report you should assign to each). Note that you may use any software you desire for your analysis, but your clustering should be undertaken in Python for this coursework. While the report is relatively flexible in structure, it is expected that you will have the following sections included:

1. **An Executive Summary:** including a description of the task, a summary of your technical approach, a summary of the data that underpins it, a summary of the results, and a summary of the insights you have arrived at.
2. **A Feature Description section:** a summary of the features you have selected and/or generated from the data to describe customers, justifying the strategy you have taken.
3. **A Customer Base Summary section:** A cursory exploratory analysis of the data you've received, summarising the company's market according to the features you have developed.
4. **A Segmentation Methodology section:** A description of the clustering approach you have taken, justifying what you think is a good value of k, the number of groups to use.
5. **A Results section:** Having produced your clusters, you must now name and summarise them. This should be done both statistically, describing how the features you have used vary across different clusters, and descriptively (each cluster should be accompanied by a single paragraph detailing a brief vignette that describes the customer 'archetype' reflected by that cluster).
6. **A Summary Section:** Here you will provide a summary of your results, the business case for your clustering solution and a recommendation for two segments that you believe will be of most importance to the company to focus attention on (including your reasons why). Draw together any key take-home points and marketing recommendations you may have as a result of your analysis, or indeed any suggestions you have to the client for further analysing business recommendations for further potential analysis.

4. Marking Criteria

Your submission will be assessed based on the following mark scheme:

- Work Plan (submitted on 20th February) (5 marks)
- Executive Summary (5 marks)
- Appropriateness of Data preparation (5 marks)
- Appropriateness of Feature Selection/Engineering (20 marks)
- Description and Justification of Methodology (10 marks)
- Analysis and Description of Results (25 marks)
  - Overall Customer base summary (5 marks)

- Individual Statistical Summaries of Clusters (10 marks)
- Pen Portraits of Clusters (10 marks)

- Efficacy of Insights and Recommendations (5 marks)
- Technical Implementation (20 marks)
  - Functionality (15 marks)
  - Clarity and Commenting (5 marks)
- Overall presentation and professionalism of Report (5 marks)

4. Submission Guidelines

Your **Work Plan** must be submitted by **3pm on Friday 20th February 2026** via email to georgiana.nica-avram1@nottingham.ac.uk using the email title "ASA Coursework 1 - Workplan", or handed to Prof Korfiatis/Prof Goulding in person at the ASA lecture the day before.

Your **Final Report** must be submitted by 3pm on 5th March via Moodle. In your submission please attach:

1. Your Final Report (Do not forget, this has a strict maximum of 3000 words, with no appendices);
2. Your technical implementation / code;
3. Your results file denoting customer IDs and their attributed segment.

As per University guidelines late submissions will lose 5% from their final mark per day. All text, code and workflows will also be examined to ensure there is no repetition between submissions. Any plagiarised work will immediately receive zero marks.

5. Example Work Plan For ASA Coursework

- Define customer variables (e.g., behavioural variables, such as prior purchases, average spend, time patterns, category, quantity)
- Cleanse data: Normalize the data
- Feature Engineering (e.g., perform Principal Component Analysis to reduce the dimensionality)
- Feature Selection
- Cluster into Segments:
  - K-means: from sklearn.cluster import KMeans
  - Cluster number evaluation: from sklearn.metrics import silhouette_score
  - Create a clusterer that fits to 6 segments
  - Cluster fit evaluation: predict the cluster for each datapoint or calculate a silhouette score for the 6 segment solution
- Analyse Segment Results:
  - Cluster Visualisation
- Creating Profiles:
  - Use groupby() and describe() to paint a picture of each cluster in the original space, compare that to your centroid results above. From these, come up with a short description of each segment that could be used by the business, before we compare them to the reality.
- Identify the two most attractive targets:
  - Evaluate attractiveness of each segment

- Select which segment to market to

6. Final Message from the company's Chief Data Officer

"Dear consultant - we are aware customer segmentation is somewhat subjective, so we are happy to offer the following guidance. The segmentations you produce will be used by ourselves to explore our customer base in the first instance, but then subsequently for marketing, and your suggestions should be guided by this usage. We also understand the job will be to decide how best to describe each customer (i.e. feature selection). While we would be very interested in new ways you can think of to describe our consumers' behaviour, the following ways of describing customer behaviour might be worth investigating:

- ***Spend:*** *It is often said that 20% of a customer base accounts for 80% of a company's profits. This may or may not be true, but it does isolate the fact that some groups will produce more revenue for the company than others, and this needs to be a part of the analysis. Hence a customer's "**total_spend**" might be included as a feature.*
- ***Frequency:*** *We have noted that some of our customers exhibit different shopping patterns over time. The frequency of visitation in a shopper's patterns could well help to underpin customer segments, so the "**number_of_visits**" may be worth considering.*
- ***Average Spend:*** *We definitely want to consider the average item spend of each group generated by the clustering process - are customers in a group purchasing many small items or single higher cost products? This will give us an indication of whether that customer type is focussed on low or high involvement products, and thus help shape our relationship with them. You may wish to use a customer's **average_spend_per_item** within your features. Along the same lines, we also think you may want to include the **average_basket_spend** and **average_item_count** for each visit as features.*
- *And finally, the ability to distinguish customers in each segment by the different types of products or categories that they buy from would allow us to potentially sculpt marketing effort to different groups based on their actual interests, preferences and shopping missions.*

Of course, these are just some *of the core items you might want to take into account when you design the features you are using to input into your clustering algorithm. Please extend as you see fit. We appreciate your efforts and understand that time constraints mean that **you won't be able to cover everything** - nonetheless we look forward to seeing your individual 'cut' through this data in the time you have available for the study." - CDO.*

END OF COURSEWORK SPECIFICATION